

Paper : SEC-A-2
(Software Engineering)
Full Marks : 80

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **question nos. 1 and 2** and **any four** questions from the rest.

1. Answer **any ten** questions :

2×10

- (a) List the goals of Software Engineering.
- (b) Write the advantages of using Prototype.
- (c) What is software testing?
- (d) What is context diagram?
- (e) List out the characteristics of good SRS.
- (f) What do you understand by bug in code?
- (g) Write the names of phases in a software development life cycle.
- (h) What is unit testing?
- (i) What is heuristic based approach?
- (j) Why is risk analysis important?
- (k) Explain the importance of maintenance phase.
- (l) What is software failure?
- (m) Write any two differences between “known risk” and “predictable risk”.
- (n) Why is LOC not the better metric to estimate the cost of a software?

2. Write short notes on (**any four**) :

5×4

- (a) Data Dictionary
- (b) Module Driver and Stub
- (c) Software Reliability and Availability
- (d) Cyclomatic complexity
- (e) Software Verification and Validation
- (f) Types of Coupling.

Z(5th Sm.)-Computer Sc.-G/SEC-A-1 & SEC-A-2/CBCS (4)

3. (a) Discuss the distinct tasks involved in requirement engineering process.
(b) Describe how software requirements are documented.
(c) State the importance of documentation. 4+3+3
4. Design Level-0 and Level-1 DFD for railway reservation system. 2+8
5. (a) Define Risk.
(b) List types of risk and give examples for each.
(c) Explain the phases of risk management. 2+5+3
6. (a) Discuss different attributes that contribute to the quality of a software.
(b) Differentiate between black box and white box testing. 6+4
7. (a) What are the advantages of iterative waterfall model over classical waterfall model?
(b) What is the significance of the spiral model compared to other life cycle model?
(c) What is the impact of reusability in software development process? 3+4+3
8. (a) What is software maintenance? Briefly describe any two types of software maintenance.
(b) Explain any one integration testing in detail. (2+4)+4
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